

Lab: Free Energy and Perceptual Inference

Learning Goal: Compute free energy and precision-weighted prediction errors for simple scenarios.

Part 1: Computing Prediction Error

Exercise: For each scenario, identify o , $g(s)$, and compute the prediction error ($o - g(s)$):

Scenario	Observation o	Prediction $g(s)$	Error
You expect the room to be 20°C but measure 24°C	24	20	
You predict your friend will arrive at 7:00 PM and they arrive at 7:00 PM	7:00	7:00	
You expect a package to weigh 1kg but it weighs 3kg	3	1	

Write your response here

Part 2: Precision Weighting

Learning Goal: See how precision changes the influence of prediction errors.

Exercise: Using the formula $F \propto \frac{1}{2} \pi(o - g(s))^2$, compute weighted prediction error for:

Scenario	Error ($o - g(s)$)	Precision π	Weighted error $\frac{1}{2}\pi(o-g(s))^2$	Interpretation
Trusting a digital thermometer	4	10		
Trusting a broken thermometer	4	0.1		
Slight color difference in bright light	2	5		
Slight color difference in dim light	2	0.5		

Which errors matter more? Why?

Write your response here

Part 3: Free Energy Decomposition

Learning Goal: Analyze the accuracy-complexity trade-off.

Scenario: Two models explain why your friend is late:

- Model A (simple): "Traffic is bad" — complexity = 1, accuracy = 5
- Model B (complex): "There was a traffic jam caused by a circus elephant that escaped and a simultaneous meteor shower" — complexity = 10, accuracy = 6
- Compute $F = \text{complexity} - \text{accuracy}$ for each (note: lower F is better)
- Which model does the free energy principle prefer? Why?

- How does this relate to Occam's razor?

Write your response here

Part 4: Hierarchical Prediction Error

Learning Goal: Trace prediction errors up and predictions down through a hierarchy.

Exercise: Consider a 3-level hierarchy for reading a word:

- Level 3 (context): Predicts the sentence topic
- Level 2 (word): Predicts the next word based on context
- Level 1 (letters): Predicts which letters based on the word

You're reading: "The cat sat on the ____" and the next word is "mat."

1. What does Level 3 predict? Is there prediction error?
2. What does Level 2 predict? Is there prediction error?
3. What does Level 1 predict? Is there prediction error?
4. Now change the word to "xylophone." Where are the prediction errors largest?

Write your response here

Part 5: Reflection

In 150 words, reflect: The equation $F \propto \frac{1}{2} \pi(o - g(s))^2$ is remarkably simple — yet it formalizes perception, attention, and belief updating. Why is mathematical simplicity valuable in a scientific theory?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Prediction error computation	Basic prediction error
2	Precision analysis	Weighted prediction errors
3	Trade-off evaluation	Accuracy-complexity balance
4	Hierarchical tracing	Multi-level prediction errors
5	Theory reflection	Value of mathematical formalization